(d) No heat loss to surrounding. [B1]

3 (a) Newton's second law of motion states that the rate of change of momentum is proportional to the net / resultant (external) force (acting on it) and the change (of momentum) takes place in the direction of the (net) force [A1]

(b) (i)

$$s = ut + \frac{1}{2}at^2 = (20)(12) + \frac{1}{2}\frac{4800}{800}(12)^2 \quad \text{[C1]}$$

$$= 672 \text{ m} \quad \text{[A1]}$$

(ii)

$$v = u + at = 20 + \frac{4800}{800}(12) \quad \text{[C1]}$$

$$= 92 \text{ m s}^{-1} \quad \text{[A1]}$$

(iii) 1.

$$\text{work} = Fs = (4800)(672) \quad \text{[M0]}$$

$$= 3.23 \times 10^6 \text{ J} \quad \text{[A1]}$$

2.

$$\text{work} = \text{gain in KE}$$

$$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}(800)(92)^2 - \frac{1}{2}(800)(20)^2 \quad \text{[M0]}$$

$$= 3.23 \times 10^6 \text{ J} \quad \text{[A1]}$$

(iv) impulse = change in momentum

= $mv - mu$ = (800)(92) – (800)(20) [C1]

= 5.76×10^4 N s [A1]

OR use impulse = Ft = (4800)(12) = 5.76×10^4 N s

4 (a)